CHAPTER 1: INTRODUCTION AND OBJECTIVE

1.1 Introduction:

Education plays a vital role in the development of society, and schools handle a large amount of information every day. Student admissions, teacher records, attendance, examination results, fee collection, notifications, complaints, and leave applications are some of the important activities performed regularly in every educational institution. In many schools, these tasks are still managed manually using registers, notebooks, and paper files. Although this traditional method has been followed for many years, it is time-consuming, difficult to maintain, and more likely to produce errors.

With the advancement of computer technology, schools are increasingly adopting computerized systems to improve their administrative work. A School Management System provides a centralized platform that stores and manages all important information digitally. It allows authorized users to perform different tasks quickly while ensuring that records remain organized and easily accessible.

The **School Management System** developed in this project is a desktop application created using the **C++ programming language**. The project mainly focuses on applying Object-Oriented Programming (OOP) concepts together with file handling to build a practical real-world application. The system provides different login panels for **Super Admin, Principal, Teacher, Student, and Parent**, where each user has different responsibilities and permissions according to their role.

The Super Admin is responsible for managing schools and principals. The Principal manages teachers and oversees school activities. Teachers can admit students, mark attendance, enter examination results, set fees, send notifications, and request student removal. Students can access their academic information, while parents can view information related to their children. This role-based structure improves security and ensures that users can only access the features assigned to them.

The application stores all information permanently using text files, allowing data to remain available even after the program is closed. The use of file handling also demonstrates practical implementation of C++ concepts such as classes, objects, functions, inheritance, encapsulation, and data management.

Overall, the School Management System simplifies school administration, reduces paperwork, improves data accuracy, saves valuable time, and provides a user-friendly solution for managing school records efficiently.

1.2 Project Profile:

The School Management System is a menu-driven desktop application designed to automate various administrative and academic activities performed in a school. The project has been developed using the C++ programming language and follows the principles of Object-Oriented Programming to make the application modular, organized, and easy to maintain.

The application provides separate login accounts for different categories of users, ensuring proper access control and data security. Each user performs only those operations that belong to their responsibilities. Data is stored using text files through file handling techniques, allowing records to be saved permanently without requiring any external database.

The project has been developed primarily for educational purposes to demonstrate the practical implementation of programming concepts learned during the course. It also provides experience in designing menu-driven applications, implementing multiple user roles, validating user input, generating unique IDs, and maintaining structured records

Project Details

- **Project Title:** School Management System
- **Project Type:** Desktop Application
- **Category:** Educational Management System
- **Programming Language:** C++
- **Programming Paradigm:** Object-Oriented Programming (OOP)
- **Development Environment:** Visual Studio Code
- **Compiler:** GNU G++ / MinGW
- **Operating System:** Windows
- **Database:** File Handling (.txt files)
- **Interface:** Console-Based Menu Driven System

1.3 Objectives:

- To reduce manual paperwork involved in school administration.
- To provide role-based access for Super Admin, Principal, Teacher, Student, and Parent.
- To maintain school information in an organized digital format.
- To generate unique IDs automatically for different users.
- To manage attendance records efficiently.
- To maintain examination results accurately.
- To store fee-related information securely.
- To provide notification and complaint management facilities.
- To minimize human errors during record management.

1.4 Existing Manual System:

In the traditional school management system, almost every activity is performed manually using registers, notebooks, files, and paperwork. Student admission forms, teacher records, attendance registers, examination results, fee receipts, and other documents are maintained physically by school staff. Although this approach is simple, it becomes increasingly difficult to manage as the number of students and employees grows.

Searching for a particular student's information often requires checking multiple registers, which consumes a considerable amount of time. Updating records manually may also result in overwriting, duplication, or missing information. Generating reports becomes difficult because every calculation must be performed manually.

Manual systems also increase the possibility of losing important documents due to physical damage, misplacement, or improper storage. Maintaining consistency among different records becomes challenging, especially when multiple staff members update information simultaneously.

Therefore, many educational institutions are replacing manual systems with computerized solutions that improve efficiency, security, and data accuracy.

1.4.1 Limitations of the Existing Manual System:

- Requires large amounts of paperwork.
- Time-consuming for maintaining records.
- Difficult to search old information.
- Duplicate records may occur.
- Records can be lost or damaged easily.
- Updating information requires significant manual effort.
- Report generation becomes difficult.
- Data consistency cannot always be maintained.
- Security of confidential information is limited.
- Managing large numbers of students becomes complicated.

1.5 Proposed System:

The proposed School Management System is designed to replace manual record management with a computerized solution developed using C++. The application automates several school activities by storing information digitally through file handling.

Different users receive separate login credentials according to their roles. The Super Admin manages schools and principals, Principals manage teachers, Teachers manage students and academic activities, Students access their information, and Parents monitor their children's records.

The system automatically generates unique IDs, validates user input, stores records permanently, and provides search, update, and delete operations. Since all information is maintained digitally, retrieving records becomes faster and more accurate.

The proposed system reduces paperwork, minimizes errors, improves security, and provides an organized method for maintaining school information efficiently.

1.5.1 Advantages of Proposed System:

- Reduces paperwork significantly.
- Saves time during daily operations.
- Improves data accuracy.
- Stores records permanently using file handling.
- Easy searching and retrieval of records.
- Faster updating and deletion of information.
- Provides role-based security.
- Generates unique IDs automatically.
- User-friendly menu-driven interface.
- Better organization of school records.
- Reduces duplicate entries.
- Easy maintenance and future enhancement.
- Demonstrates practical implementation of C++ concepts.

CHAPTER 2: REQUIREMENT ANALYSIS

2.1 Introduction:

Requirement Analysis is one of the most important stages of software development. During this phase, the functional and non-functional requirements of the project are identified and analyzed before implementation begins. Proper requirement analysis helps developers understand the resources, software, hardware, and user requirements needed for the successful completion of the project.

The School Management System has been designed to satisfy the requirements of educational institutions by providing an easy-to-use, secure, and efficient application for managing different school activities.

2.2 Software Requirements:

- ✓ **Operating System:** Windows.
- ✓ **Programming Language:** C++.
- ✓ **Code Editor:** Visual Studio Code.
- ✓ **Compiler:** C++ Compiler.
- ✓ **Storage:** Local files for storing records.

2.3 Hardware Requirement:

- ✓ Minimum 4GB RAM
- ✓ Dual-core processor.
- ✓ Compatible with desktops or laptops.

2.4 Training Requirement:

Basic understanding of:

- Computer operation.
- Menu-based applications.
- Different user roles and their functions.
- Basic C++ knowledge for maintaining the program.

2.5 Descriptions:

The School Management System consists of multiple modules that work together to perform different administrative and academic operations.

- **Super Admin:** Creates schools, manages principals, searches schools, edits school information, and removes records.
- **Principal:** Manages teachers belonging to the assigned school and monitors school-related activities.
- **Teacher:** Admits students, records attendance, enters examination results, updates marks, manages fees, sends notifications, and submits removal requests.
- **Student:** Views attendance, examination results, fee details, notifications, and personal information.
- **Parent:** Accesses information related to their child, including attendance, results, notifications, and fee records.

The interaction among these modules provides an organized management system for the entire school.

2.6 Why C++?

C++ has been selected because it is one of the most powerful programming languages for learning Object-Oriented Programming and system development.

Advantages include:

- Supports Object-Oriented Programming.
- High execution speed.
- Efficient memory management.
- Powerful file handling capabilities.
- Supports classes and inheritance.
- Suitable for menu-driven applications.
- Easy to implement modular programming.
- Widely used in academic projects.
- Helps strengthen programming fundamentals.

CHAPTER 3: FEASIBILITY STUDY

3.1 Introduction:

A feasibility study is performed before developing a software project to determine whether the project is practical, economical, and beneficial. It evaluates different factors such as technical resources, financial requirements, operational efficiency, and implementation possibilities.

For the School Management System, the feasibility study confirms that the project can be successfully developed using available software and hardware resources without significant cost.

3.2 Technical Feasibility:

Technical feasibility determines whether the required technology, hardware, and software are available for developing the application.

The School Management System is technically feasible because it uses C++, a standard programming language supported by most operating systems and compilers. The project requires only a normal computer, Visual Studio Code, and a C++ compiler. File handling eliminates the need for an external database, making development simpler while still providing permanent storage.

The application is lightweight, requires minimal system resources, and can run efficiently on ordinary desktop and laptop computers.

3.3 Economical Feasibility:

Economic feasibility determines whether the project can be completed within an affordable budget.

The School Management System requires no expensive software licenses or hardware. The compiler, development environment, and required programming tools are freely available. Since the project uses text files instead of commercial database software, the overall development cost remains very low.

Maintenance costs are also minimal because the application can be updated without additional licensing expenses.

3.4 Operational Feasibility:

Operational feasibility evaluates whether users can operate the application successfully.

The School Management System has been developed using a simple menu-driven interface that is easy to understand. Different users receive separate login panels according to their responsibilities, reducing confusion and improving security.

Since the application automates several manual activities, school staff can perform their daily work more efficiently. The system reduces paperwork, minimizes errors, and allows faster retrieval of information.

3.5 Conclusion:

The feasibility study indicates that the School Management System is technically, economically, and operationally feasible. The project can be developed using available resources without high costs while providing significant improvements over the traditional manual system.

The system offers faster data management, improved accuracy, enhanced security, organized record keeping, and better overall efficiency. Therefore, implementing this computerized School Management System is a practical and beneficial solution for educational institutions.